

WEEK 14: PENETRATION TESTING

1. Python for Penetration Testing

Python is one of the most widely used programming languages in cybersecurity and penetration testing due to its simplicity, flexibility, and powerful libraries. It allows security professionals to automate tasks, develop custom security tools, and perform efficient testing of systems and networks. This practical focused on understanding how Python can be used for penetration testing activities in a controlled lab environment as per the cybersecurity syllabus.

2. Shell Scripting

Shell scripting is a fundamental skill in cybersecurity and system administration. It involves writing scripts using a shell (such as Bash) to automate tasks, manage systems, and perform security-related operations. In cybersecurity, shell scripts are widely used to automate reconnaissance, monitoring, log analysis, and routine administrative tasks. This practical focused on understanding shell scripting concepts and implementing basic scripts in a controlled lab environment.

- Bash Shell
- Nano Editor

3. DoS and DDoS Attacks